



Acoustic Detection of UAV Abnormality Using One Ground-Based Acoustic Vector Sensor

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Abstract

Defective unmanned aerial vehicles (UAVs), like other malfunctioning machinery, often emit abnormal sounds. Early detection of such acoustic abnormalities enables timely countermeasures to prevent equipment loss and human casualties. This paper proposes a deep learning scheme to passively detect abnormal sounds emitted from UAVs, using only a single ground-based acoustic vector sensor (AVS). This is because lone AVS measured sound intensity spectrogram can already provide information on both the sound source's polar-azimuthal direction and abnormality time-frequency signature. This paper integrates a "convolutional autoencoder" with an "attention module", for UAV abnormality detection. Field experiments, with a UAV in flight and a single AVS on the ground, validate the effectiveness of the proposed method, achieving an F1-score (a machine-learning accuracy metric) exceeding 95% for successful anomaly detection.

Index Terms: Acoustic vector sensor, sound intensity, UAV, anomaly detection, autoencoder, deep learning

1. Introduction

Unmanned aerial vehicles (UAVs) are compact in size, highly maneuverable, low in cost, and easy to operate. Thus, they are often used for applications in military reconnaissance, aerial photography, disaster detection, and delivery services [1–6]. Such UAVs are liable to electromechanical malfunctions, leading to mission failure, loss of equipment, and collateral damages [7]. Nonetheless, abnormal sounds from the UAV often accompany such malfunctioning. Early detection of UAV abnormalities can allow timely mitigation, so it is practically vital to develop acoustic technology to detect abnormal sounds emitted from UAVs.

Anomaly detection by sound/noise/vibration signals has frequently been applied in practice due to the advantages of non-contact measurements and long-range detection.

Unlike all existing methods that rely solely on sound pressure scalar signals for anomaly detection, the proposed approach utilizes the directional information naturally embedded in the measurements at a single acoustic vector sensor (AVS). This helps suppress isotropic environmental noise, highlighting the characteristics of anomalous UAV sounds and improving

the accuracy of anomaly detection. Existing anomaly detection methods based on scalar sound pressure can generally be categorized into model-driven and data-driven approaches.

For UAV fault detection: This paper furthermore advances the first unsupervised data-driven learning, in contrast to the supervised data-driven learning in [8–15] and the self-supervised data-driven learning in [16].

Specifically, this paper proposes an unsupervised method for UAV anomaly detection based on acoustic vector signals, using a single ground-based AVS passively receives and processes UAV noise to identify anomalies. Unlike the sound pressure signals measured by the conventional microphones, the acoustic vector signals measured by a single AVS inherently carry the information of source direction without requiring complex microphone arrays of notable spatial dimensions. It is precisely the directional information carried by the acoustic vector signals that enables effective suppression of isotropic environmental noise and highlights the characteristics of anomalous UAV sounds by directly extracting the sound intensity spectrogram that combines sound pressure and particle velocity phase information, without the need to perform complex signal enhancement algorithms based on the explicit sound source direction. The proposed method uses an unsupervised learning strategy, avoiding the need for a large amount of labeled anomalous data. An encoder-decoder-encoder architecture is adopted, which introduces the reconstruction error of the encoded features into the loss function, making it more effective at capturing anomaly patterns in the spectrogram. Additionally, a Convolutional Block Attention Module (CBAM) [17] was integrated into the encoder, which dynamically assigns varying weights to salient and less significant features during the learning process, emphasizing crucial features to improve detection accuracy.

2. Data Acquisition

Consider a UAV inflight in the far field above an acoustic vector sensor (AVS) mounted on the ground outdoor. The UAV emits a sound of $s(t)$, which impinges upon the AVS at a polar-azimuthal direction of (θ, ϕ) . The AVS' measured data at the n th time instant is expressible as

$$\mathbf{y}(n) = \begin{bmatrix} p(n) \\ \mathbf{v}(n) \end{bmatrix} = \begin{bmatrix} 1 \\ \mathbf{u} \end{bmatrix} s(n) + \begin{bmatrix} e_p(n) \\ \mathbf{e}_v(n) \end{bmatrix}, \quad (1)$$

where $p(n)$ symbolizes the pressure sensor's measurement and $\mathbf{v}(n) = [v_x(n), v_y(n), v_z(n)]^T$ that of the particle-velocity sensors. Similarly, $e_p(n)$ and $\mathbf{e}_v(n) = [e_{v_x}(n), e_{v_y}(n), e_{v_z}(n)]^T$ denote the additive noise. The term

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$s(n)$ refers to the noise signal generated by the hovering UAV, whereas $\mathbf{u} = [\cos \phi \sin \theta, \sin \phi \sin \theta, \cos \theta]^T$ represents the unitary direction vector pointing from the AVS towards the UAV. This measured sound data $\{\mathbf{y}(n), \forall n\}$ will be utilized to diagnose the UAV for any potential abnormalities.

A field experiment was carried out to mimic the above scenario: The UAV was an SYMA X30SE quadcopter; please see Figure 1a. It hovered 3 meters aboveground, but 4 meters of groundlevel distance from an AVS placed on a tripod 1 meter aboveground in a grassy sports field in the Sichuan University campus; please see Figure 1b. The AVS was model WL-AVS-3D, distributed by Microfluidic Sound Sensation. This AVS had four channels: 1 pressure sensor plus 3 particle-velocity sensors.

Around the receiver, the UAV hovered with 6 horizontal directions at azimuth angles of $\phi = \{0^\circ, 45^\circ, 90^\circ, 180^\circ, 225^\circ, 270^\circ\}$.

Onboard the above-described UAV inflight, two types of abnormal sound was emitted: (a) sound from an online video of an UAV with unknown motor faults and broadcasted by a Bluetooth loudspeaker mounted on the UAV, with adjustable volume to accommodate different signal-to-noise ratios (SNRs);¹ (b) The natural noise emitted by the UAV after two of its four propeller blades were intentionally damaged.

In contrast, the acoustic signals collected during the UAV's normal flight — i.e., with intact propeller blades and no external sound sources — are considered the normal sounds in this study.

The AVS was connected to a National Instruments 9234 data acquisition card, with the sampling frequency of 51.2 kHz. The captured data are processed by National Instruments LabVIEW software on a laptop.

The AVS' data were bandpass-filtered to pass frequencies from 0.2 to 6.6 kHz. Thus, two-second data samples were formed as 4×102400 matrices. Each data sample was assigned a binary label of either "normal" or "abnormal".

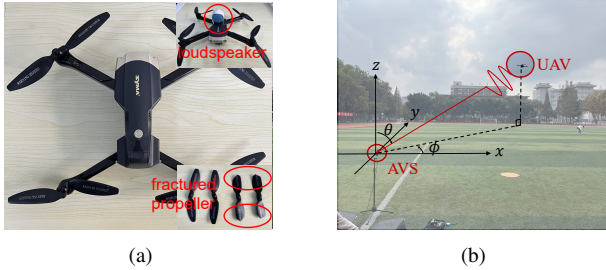


Figure 1: (a) The UAV employed in the field experiment, with fractured propeller blades to emit abnormal sounds while in-flight. (b) The grassy field in Sichuan University where the field experiment was conducted.

3. Proposed Method

3.1. Feature extraction

The 4×1 vector samples measured by the AVS were grouped into frames, and each channel's data then individually underwent short-time Fourier transformation (STFT) on a moving

time-window.

At the m -th time-frame and the k -th frequency-bin, denote the pressure signal as $P(k, m)$ and the three particle-velocity signals as $V_x(k, m)$, $V_y(k, m)$, and $V_z(k, m)$.

For an emitter impinging from the far field of the AVS, the sound pressure and particle velocity are phase-aligned [18], and their cross-spectrogram energy is concentrated in the real part [19], that is,

$$\begin{cases} I_x(k, m) = \Re \{P(k, m)V_x^*(k, m)\}, \\ I_y(k, m) = \Re \{P(k, m)V_y^*(k, m)\}, \\ I_z(k, m) = \Re \{P(k, m)V_z^*(k, m)\}. \end{cases} \quad (2)$$

The magnitude of sound intensity equals

$$I(k, m) = \sqrt{I_x^2(k, m) + I_y^2(k, m) + I_z^2(k, m)}, \quad (3)$$

which corresponds to the UAV's direction relative to the AVS.

By collecting all discrete values $I(k, m)$ over time frames $m = 1, \dots, M$ and frequency bins $k = 1, \dots, K$ (spanning the frequency range of 0.2 kHz to 6.6 kHz), the sound intensity spectrogram for a single data sample is represented as a matrix $\mathbf{I} \in \mathbb{R}^{K \times M}$. Segment $\mathbf{I}$ into four submatrices $\mathbf{F}_i \in \mathbb{R}^{K \times M'}$, $i \in \{1, 2, 3, 4\}$ by using a time-domain sliding window, each representing a $K \times M'$ feature map.

On the other hand, the pressure sensor omnidirectionally measures the sound field as a scalar signal $P(k, m)$, regardless of the UAV's direction. Figure 2 compares the sound intensity spectrogram with the sound pressure spectrogram for an identical data sample, visually demonstrating that the former better highlights the features of UAV noise than the latter does.

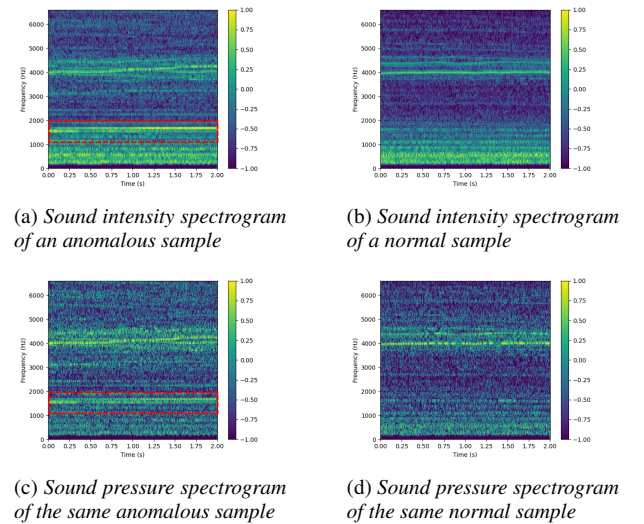


Figure 2: The red-boxed regions in (a) and (c) represent the anomalous features in the sound intensity spectrogram and sound pressure spectrogram, respectively. By comparing them with the normal spectrograms in (b) and (d), it can be observed that the sound intensity spectrogram better highlights the anomalous acoustic characteristics of the UAV.

¹<https://v.douyin.com/l5dOjDxM7jw>

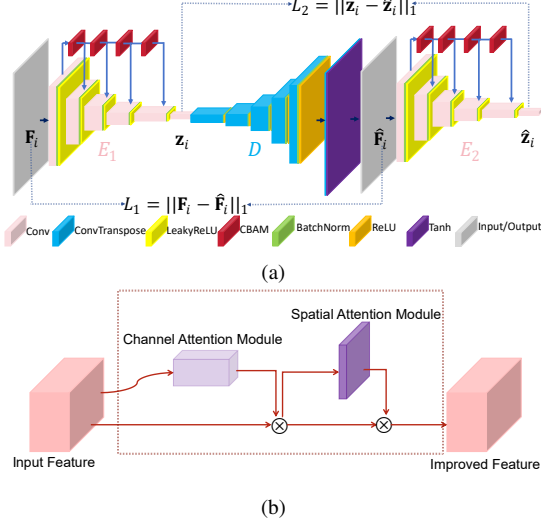


Figure 3: (a) The network structure of the proposed convolutional autoencoder with the CBAM. (b) The structure of CBAM.

3.2. Anomaly detection model

3.2.1. Model structure

The proposed anomaly detection model in Figure 3 is inspired by a generative adversarial network specifically designed to implement anomaly detection, i.e., the GANomaly in [20]. In the proposed design, the GANomaly's generator with the encoder-decoder-network structure is adopted, while the GANomaly's discriminator is abandoned, aiming to lower the computational complexity. Additionally, a sequence of CBAMs [17] are incorporated in both encoders E_1 and E_2 , as shown in Figure 3(a), aiming to capture the global feature through the channel and spatial attentions. The first encoder E_1 maps the input feature map F_i to a latent representation z_i , a d dimensional vector. The decoder D reconstructs the feature map $\hat{F}_i$ from z_i . Similarly to E_1 , the second encoder E_2 maps $\hat{F}_i$ to another d dimensional vector $\hat{z}_i$, which is supposed to be an estimate of z_i .

The above designed network is fundamentally a convolutional autoencoder with the CBAMs, or CAE-CBAM in brief.

3.2.2. Model training

The proposed CAE-CBAM in Section 3.2.1 is trained exclusively with the normal samples obtained in Section 2, with the following loss function [20]:

$$L = \omega_1 \underbrace{\|F_i - \hat{F}_i\|_1}_{:=L_1} + \omega_2 \underbrace{\|z_i - \hat{z}_i\|_1}_{:=L_2}, \quad (4)$$

where ω_1 and ω_2 are empirically selected to weigh L_1 and L_2 , and $\|\cdot\|_1$ indicates the ℓ_1 norm.

For the well trained CAE-CBAM solely on the normal samples, the encoder E_1 extracts the latent representation z_i of the normal feature map F_i , while the decoder D reconstructs the estimate $\hat{F}_i$, and the encoder E_2 reconstructs the estimate $\hat{z}_i$. The proposed model tends to minimize L in (4). On the contrary, if an abnormal feature map is fed into the model, L tends to significantly rise.

3.2.3. Anomaly score

The anomaly score $\mathcal{A}(F_i)$ of each F_i can be defined as the ℓ_1 distance between z and $\hat{z}$ [21], i.e., $\mathcal{A}(F_i) = L_2 = \|z - \hat{z}\|_1$. Thus, the anomaly score for sample I can be obtained by taking the average as $\mathcal{A}(I) = \frac{1}{4} \sum_{i=1}^4 \mathcal{A}(F_i)$.

The anomaly detector can be thus established as

$$\mathcal{H}(I, \lambda) = \begin{cases} 0 & (\text{Normal}) & \mathcal{A}(I) \leq \lambda \\ 1 & (\text{Abnormal}) & \mathcal{A}(I) > \lambda \end{cases}, \quad (5)$$

where λ is a preset threshold, which can be determined by maximizing Youden's statistics [22].

4. Real Data Validation

4.1. Dataset construction

With the data samples obtained in Section 2, 2042 normal samples are randomly selected for model training. The validation set and the testing set are formed by randomly choosing from the remaining samples. Two types of the UAV's motor fault sounds played through the Bluetooth speakers with the different SNR² are considered. A practical type of abnormal sound due to the UAV's propeller blade damage is also considered. Please see Table 1.

Table 1: Dataset partitioning

	Training	Validation	Testing
Normal samples	2042	128	122
Abnormal samples (motor fault, SNR = -2.6 dB)	0	192	199
Abnormal samples (motor fault, SNR = -5.3 dB)	0	0	243
Abnormal samples (propeller blade damage)	0	0	292

4.2. Hyperparameters

The proposed CAE-CBAM model adopts the Adam optimizer [23], with an initial learning rate $r = 2 \times 10^{-4}$, momentum parameters $\beta_1 = 0.1$, and $\beta_2 = 0.999$. The training process is conducted over 20 epochs, employing the loss function L in equation (4), where the weights are empirically set as $\omega_1 = 100$ and $\omega_2 = 1$ for the optimum results. Each of the encoders E_1 , E_2 , and the decoder D consists of 5 convolutional layers.

4.3. Performance evaluation

To prevent overfitting, the model's performance is evaluated on the validation set after each epoch, and the best-performing model parameters are saved. Anomaly detection is performed with on a threshold $\lambda = 0.03548$ in equation (5). Figure 4 illustrates the distribution of anomaly scores for the testing set, where the left-side histogram colored in blue represents the anomaly score distribution for only the normal samples, and

²In the SNR calculation, the played motor fault sound is regarded as the target signal, while the UAV's own noise and the ambient environmental noise are collectively treated as the background noise.

the right-side histogram colored in pink represents the anomaly score distribution for only the abnormal samples.

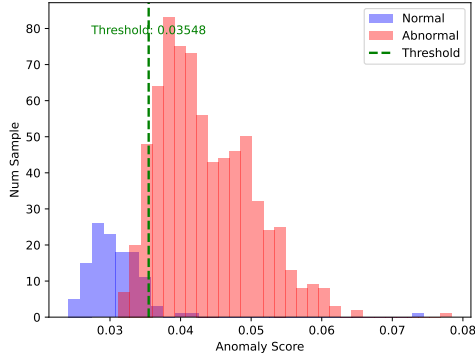


Figure 4: The histogram of anomaly scores for normal and abnormal samples in the testing set (as described in Table 1) with the decision threshold $\lambda = 0.03548$.

To assess the performance of the proposed CAE-CBAM model, the metrics accuracy, precision, recall, and F1-score are considered. To validate the superiority of anomaly detection using acoustic vector signals, the performance of CAE-CBAM with the input sound intensity spectrograms is compared to that with the input sound pressure spectrograms. Table 2 demonstrates that the proposed CAE-CBAM performs well on the testing set, for both types of feature maps of the intensity and pressure spectrograms. Moreover, the intensity spectrograms show a superior performance over the pressure spectrograms, by increasing accuracy, precision, recall, and F1-score of 10.09%, 0.95%, 12.26%, and 7.39%, respectively.

Table 2: Comparison of the CAE-CBAM’s anomaly detection performance on the testing set.

Spectrogram	Accuracy	Precision	Recall	F1-Score
Intensity	92.87%	99.12%	92.51%	95.70%
Pressure	81.78%	98.17%	80.25%	88.31%

In anomaly detection tasks, minimizing false negatives is critical for ensuring the robustness of the system. Table 3 demonstrates that the proposed CAE-CBAM has a superior performance with the intensity spectrogram over the pressure spectrogram in detecting motor faults and propeller blades damage, with improvements in recall rates of 0.5%, 14.00%, and 12.68%, respectively. Notably, the motor fault sound characterized by a lower SNR, and the abnormal sound associated with propeller damage are difficult to be distinguished by human hearing. The substantial improvements in recall for these two challenging cases indicate that CAE-CBAM possesses a stronger capability in identifying anomalies under low-SNR conditions.

5. Conclusion

This paper proposes an unsupervised method for UAV anomaly detection based on acoustic vector signals, developing the an

Table 3: Comparison of recall for different types of anomalies by using intensity and pressure spectrograms.

Spectrogram	Motor fault (SNR = -2.6 dB)	Motor fault (SNR = -5.3 dB)	Propeller blade damage
Intensity	98.99%	95.06%	85.96%
Pressure	98.49%	81.06%	73.28%

encoder-decoder-encoder structure autoencoder integrated with CBAMs. Field experiments confirm the efficacy of the proposed approach, highlighting the significant advantages of acoustic vector signals over the sound pressure signals in UAV anomaly detection, particularly in detecting low-SNR anomalies and propeller faults. The directional information provided by acoustic vector signals highlights the characteristics of anomalous UAV sounds, significantly improving the detectability of such sounds.

6. References

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